

Venkata Sandeep Macha

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Professional Summary

Software Engineer with 2+ years building **production** backend systems, **ETL pipelines**, and **AI-powered** applications using **Python**, **Java**, **TypeScript**, and **AWS**. Experienced across **data engineering**, **LLM integration**, **microservices**, and **cloud infra**.

Education

University of Florida

Master of Science in Computer and Information Science and Engineering

Aug 2024 – May 2026

GPA: 3.76/4.00

Sree Vidyanikethan Engineering College

Bachelor of Technology in Computer Science Engineering

Oct 2020 – Jun 2024

GPA: 9.35/10

Technical Skills

- **Languages:** Python, Java, TypeScript, JavaScript, SQL, Swift, Go, C
- **Data Engineering:** ETL pipelines, Apache Spark, AWS Glue, AWS Redshift, AWS Athena, AWS EMR, Snowflake, dbt, Airflow, Kafka, Hadoop, data modeling
- **Backend & APIs:** Spring Boot, FastAPI, REST APIs, microservices, OAuth 2.0, Docker, Kubernetes, CI/CD, AWS EC2, AWS S3, AWS SQS, AWS CloudWatch
- **AI & ML:** LangChain, Hugging Face, Semantic Kernel, TensorFlow, PyTorch, Scikit-learn, spaCy, NLTK, RAG, prompt engineering, LLM APIs
- **Databases:** PostgreSQL, MongoDB, Snowflake, AWS Redshift, DynamoDB, SQL query optimization

Professional Experience

Software Engineer

Sep 2024 – Present

IFAS, UF

- Built ML data workflows in Python with spaCy, NLTK, and Scikit-learn on AWS EC2, backed by AWS Athena SQL analytics, improving training efficiency by **25%** across 4 datasets.
- Engineered Java Spring Boot microservices exposing REST APIs across 15+ endpoints, serving 3 research teams and cutting data retrieval time by **85%** with zero post-deploy incidents.
- Built a LangChain RAG pipeline with embedding retrieval over PostgreSQL, reducing hallucination rate by **42%** across 3 retrieval configs and 15+ typed endpoints.
- Developed TypeScript frontend workflows connected to backend microservices and prototyped a Swift data viewer, improving task completion by **25%** with secure coding practices.
- Instrumented observability with AWS CloudWatch telemetry and AWS X-Ray tracing, enforcing SLOs with error budgets that cut pipeline failures by **35%** across 10+ production jobs.

Cloud Virtual Intern

Jan 2024 – May 2024

APSCHE – AWS Academy Program

- Architected ETL pipelines with AWS Glue and SQL loading into AWS Redshift, improving throughput by **37%** and cutting query latency by **26%** for 2 data teams.
- Automated batch workflows with Apache Spark on AWS EMR, processing 10+ datasets and reducing batch processing time by **33%** across 2 analytics teams.
- Deployed RESTful APIs on AWS EC2 with GitHub Actions CI/CD pipelines, reducing backend response time by **40%** and achieving zero failed deployments over a 2-week window.
- Built TypeScript connector clients with OAuth 2.0 auth handling pagination, rate limiting, and webhooks across 15+ endpoints for 2 downstream teams.

AI/ML Intern

Sep 2023 – Nov 2023

APSCHE – AWS Academy Program

- Built a retail chatbot using spaCy and NLTK for NLP intent classification on real-time AWS pipelines, improving customer interaction efficiency by **35%** across 3 product categories.
- Implemented anomaly detection using Scikit-learn and TensorFlow within structured data pipelines on AWS SageMaker, reducing diagnostic anomalies by **20%** across 4 datasets.
- Developed AI pipelines with Hugging Face transformers and LangChain agents to process 5,000+ records, cutting manual labeling effort by **45%** across 4 classification tasks.
- Containerized ML workloads with Docker and orchestrated batch inference on Kubernetes, achieving zero failed deployments across 3 model versions.

Projects

AI-Powered Healthcare Data Platform with Intelligent Document Analysis

Jun 2025 – Present

Python, Java, SQL, dbt, Snowflake, LangChain, LLM APIs, Azure OpenAI, Spark, AWS EMR, Docker

- Built ETL pipelines via dbt to process 100GB+ medical imaging and clinical document datasets, loading structured outputs into Snowflake for analytics across 3 research teams.
- Designed AI document understanding workflows using LangChain and LLM APIs via Azure OpenAI with prompt engineering to extract structured fields from 500+ unstructured clinical records, reducing manual review time by **50%**.
- Automated distributed preprocessing with Apache Spark on AWS EMR and containerized the inference stack with Docker, cutting pipeline processing time by **40%** across 8 stages with 99.9% uptime.

Certifications

AWS Solutions Architect, **AWS | AWS Associate Data Engineer**, **AWS | SQL Certification**, Coursera
Java, HackerRank | **Introduction to Cybersecurity**, Cisco